The following may be the penalties due to the occurrence of these errors:

If type I error occurs, then the company may spend towards advertisement. It may increase the expenditure of the company. On the other hand, if type II error occurs, then the company will not spend towards advertisement. It may not improve the sales of the company.

1.6 LEVEL OF SIGNIFICANCE, CRITICAL REGION AND CRITICAL VALUE(S)

In a given hypotheses testing problem, the *maximum probability* with which we would be willing to tolerate the occurrence of type I error is called **level of significance** of the test. This probability is usually denoted by 'α'. Level of significance is specified before samples are drawn to test the hypothesis.

The level of significance normally chosen in every hypotheses testing problem is 0.05 (5%) or 0.01 (1%). If, for example, the level of significance is chosen as 5%, then it means that among the 100 decisions of rejecting the null hypothesis based on 100 random samples, maximum of 5 of among them would be wrong. It is emphasized that the 100 random samples are drawn under identical and independent conditions. That is, the null hypothesis H_0 is rejected wrongly based on 5% samples when H_0 is actually true. We are about 95% confident that we made the right decision of rejecting H_0.

Critical region in a hypotheses testing problem is a subset of the sample space whose elements lead to rejection of H_0. Hence, its elements have the dimension as that of the sample size, say, $n(n > 1)$. That is,

$$\text{Critical Region} = \left\{\underset{\sim}{x} = (x_1, x_2, ..., x_n) \,|\, H_0 \text{ is rejected}\right\}.$$

A subset of the sample space whose elements does not lead to rejection of H_0 may be termed as **acceptance region**, which is the complement of the critical region. Thus,

$$\mathbf{S} = \{\text{Critical Region}\} \cup \{\text{Acceptance Region}\}.$$

Test statistic, a function of statistic(s) and the known value(s) of the underlying parameter(s), is used to make decision on H_0. Consider a hypotheses testing problem, which uses a **test statistic** $t(\underset{\sim}{X})$ and a constant c for deciding on H_0. Suppose that H_0 is rejected, when $t(\underset{\sim}{x}) > c$. It is to be noted here that $t(\underset{\sim}{X})$ is a scalar and is of dimension one. Its sampling distribution is a univariate probability distribution. The values of $t(\underset{\sim}{X})$ satisfying the condition $t(\underset{\sim}{x}) > c$ will identify the samples in the sample space, which lead to rejection of H_0. It does not mean that $\left\{t \mid t(\underset{\sim}{x}) > c\right\}$ is the corresponding critical region. The value 'c', distinguishing the elements of the critical region and the acceptance region, is referred to as **critical value**. There may be one or many critical values for a hypotheses testing problem. The critical values are determined from the sampling distribution of the respective test statistic under H_0.

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